



MAKERERE UNIVERSITY

DATA ANALYTICS & VISUALIZATION TAKE-HOME EXAMINATION

1. Examination Overview and Purpose

This take-home examination is designed to assess your practical competence in the end-to-end data analysis pipeline, from raw data acquisition through exploration, cleaning, visualization, and statistical reasoning to written communication of findings. Each student has been assigned a **unique real-world dataset** drawn from reputable public repositories. You are expected to work **independently**. All analysis must be performed using either **Python** (recommended: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, Scikit-learn) or **R** (recommended: tidyverse, ggplot2, dplyr, tidyr, caret / tidymodels) and Tableau for dashboards.

The examination covers 4 assessed parts worth a combined **100 marks**, plus an optional bonus task worth up to **5 additional marks**. **Deadline for submission is 15th July 2026 at 11:59pm, there will not be any extensions after this**. Submit to this form (<https://forms.gle/2FUqs53h7LWH8Zvg9>)

1.1 Marks Distribution at a Glance

Part A	Data Exploration & Quality Assessment	20
Part B	Data Cleaning & Preprocessing	20
Part C	Data Visualization & Analysis	35
Part D	Report & Code Quality	25
Bonus	Interactive Dashboard (optional)	+5
TOTAL		100

1.2 Learning Outcomes Assessed

Upon successful completion of this examination, students will demonstrate the ability to:

- Load, inspect, and critically evaluate the quality of a real-world dataset.
- Apply appropriate data cleaning and feature engineering techniques, with justification.
- Design and produce a comprehensive suite of statistical visualizations using correct chart types.
- Conduct and interpret inferential statistical tests relevant to their data.
- Communicate analytical findings clearly in a structured written report.
- Write clean, reproducible, well-documented code in Python or R.

2. General Instructions and Rules

2.1 Academic Integrity

- You must work entirely independently. You may consult textbooks, course notes, and online documentation (e.g., Pandas docs, ggplot2 docs) but you must not share code, outputs, or analysis with classmates.
- You must cite all external data sources, packages, and significant references used.

2.2 Dataset Acquisition

- **Your assigned dataset is listed in Section 4 of this document. Locate your student number in the assignment table.**
- Download your dataset directly from the URL or source listed. All datasets are freely and publicly available.
- If a dataset has multiple versions or sub-files (e.g., red wine vs. white wine), you may use all files together or select one sub-file provided you state your choice clearly.
- Do not switch to a different dataset without written approval from the course lecturer.
- If a link is broken or a dataset is temporarily unavailable, contact the lecturer within the first 48 hours of the examination period.

2.3 Technical Requirements

- Use Python or R for all analysis.
- Your code must run from top to bottom without errors on a fresh kernel / R session. Test this before submitting.
- All code must be contained in a single Jupyter Notebook (.ipynb) or R Markdown (.Rmd) file with logical section headers matching the parts of this exam.
- Figures must be embedded in the notebook output AND in the final report document.

2.4 Submission Requirements

Submit all of the following items via the submission form(<https://forms.gle/12sURvp7Ru5D934b9>) as a single .zip archive named:

NAME_STUDENTNUMBER_DATASET_NO.zip

3. Examination Tasks

Complete all parts below using your assigned dataset. Each part builds on the previous one. Read all instructions for a part before starting it. Where a task asks you to "justify" a decision, write your reasoning as a code comment or a Markdown cell immediately preceding or following the relevant code. All visuals in the report should have an insight explaining them.

Part A: Data Exploration and Quality Assessment

This part establishes your foundational understanding of the dataset before any modifications are made. Perform all tasks on the raw, unmodified data.

A1. Initial Data Loading and Structural Inspection

1. Load the dataset into a Pandas DataFrame or R tibble and display the first 10 rows.
2. Report the dataset dimensions (number of rows and columns).
3. List all column names alongside their data types. Identify which columns are: numerical (continuous or discrete), categorical (nominal or ordinal), datetime, or text-based.
4. Display a concise summary of the dataset (e.g., `df.info()` or `glimpse()`).

A2. Descriptive Statistics

5. Compute summary statistics for all numerical columns: count, mean, median, standard deviation, minimum, 25th percentile, 75th percentile, and maximum.
6. For categorical columns, report the number of unique categories and the top-3 most frequent values with their counts and percentages.
7. Identify the target or response variable in your dataset (if applicable) and describe its distribution.

A3. Missing Value Analysis

8. Count the total number of missing values per column and express each as a percentage of total records.
9. Visualize the missing value pattern using a heatmap, bar chart, or missingno matrix.
10. Classify columns by missingness severity: None (0%), Low (<5%), Moderate (5-20%), or High (>20%).
11. Hypothesize whether the missing data is likely MCAR (Missing Completely At Random), MAR (Missing At Random), or MNAR (Missing Not At Random), and briefly explain your reasoning.

A4. Duplicate Detection and Consistency Checks

12. Check for exact duplicate rows. Report how many were found.
13. Identify any apparent inconsistencies in at least two columns (e.g., negative ages, impossible dates, invalid categorical entries, mixed-case strings for the same category).
14. For any numerical column that represents a constrained quantity (e.g., percentage, rating, count), verify that all values fall within the valid range.

A5. Outlier Detection

15. Select at least two numerical columns and apply both the Interquartile Range (IQR) method and Z-score method to detect outliers. Report the count of outliers found by each method.
16. Visualize the distribution and outliers for each selected column using a boxplot and a histogram side by side.
17. Write a narrative summarizing your initial impressions of the dataset: what it represents, what its overall quality is, and which issues will require the most attention in Part B.

Part B: Data Cleaning and Preprocessing

Apply all cleaning and transformation steps in a clear, logical order. Create a cleaned copy of your DataFrame so the raw data is preserved. Each decision must be justified.

B1. Handling Missing Values (if-any)

18. For each column with missing values, choose and apply one of the following strategies, with written justification for each choice:

- Mean or median imputation (numerical columns with low to moderate missingness)
- Mode imputation (categorical columns with low missingness)
- Forward fill or backward fill (time-ordered data)
- Predictive imputation using a simple model (advanced; optional)
- Row or column removal (if missingness is too high to impute reliably)

19. After imputation, confirm that no missing values remain (or document those intentionally retained).

B2. Handling Duplicates and Inconsistencies (if-any)

20. Remove any exact duplicate rows found in Part A.

21. Resolve all string inconsistencies identified (e.g., standardize case, strip whitespace, harmonize category labels).

22. Correct or remove any records with invalid values identified in A4.

B3. Outlier Treatment (if-any)

23. For the outliers identified in A5, decide on a treatment strategy: remove, cap/clip (Winsorization), transform, or retain. Apply the chosen strategy and justify your decision in a Markdown cell.

24. Show before-and-after boxplots to illustrate the effect of your treatment on at least one column.

B4. Encoding and Type Conversion

25. Convert any columns stored in an incorrect data type to the appropriate type (e.g., strings that represent dates, numbers stored as text).

26. Apply one-hot encoding to at least one nominal categorical variable.

27. Apply label/ordinal encoding to at least one ordinal categorical variable (if present; otherwise apply it to another nominal variable and justify).

B5. Feature Engineering

28. Create at least two new derived features from existing columns. Examples: ratios, differences, interaction terms, binned age groups, time-since calculations, or text-length features. Justify why each derived feature could be analytically useful.

Part C: Data Visualization

This is the highest-weighted part of the examination. Every visualization must include: a descriptive **title**, labelled **axes** (with units where applicable), a **legend** (where multiple categories are shown), and a **2-3 sentence interpretation** in a Markdown cell immediately below the plot. Visualizations without interpretations will not receive full marks. **NB:** Choose chart types that are appropriate to the data type and the question being answered. Producing a chart of the wrong type (e.g., a pie chart with 20 categories, or a line chart for unordered categorical data) will be penalised even if it is technically correct.

C1. Univariate Analysis

29. Histogram: Plot the distribution of two different numerical variables. Use appropriate bin sizes and overlay a KDE (kernel density estimate) curve where possible. Comment on the shape of each distribution (skewness, modality).
30. Boxplot: For the same two numerical variables, produce side-by-side boxplots grouped by a relevant categorical variable. Discuss what the interquartile ranges reveal.
31. Bar Chart: For one categorical variable with 3-15 categories, plot the frequency or proportion of each category. Sort bars by descending frequency. Discuss which category dominates and whether this is expected.
32. Count Plot with Percentage Annotations: Select one binary or low-cardinality categorical column and annotate each bar with its percentage of the total.

C2. Bivariate Analysis

33. Scatter Plot: Visualize the relationship between two numerical variables. Annotate with a linear trend line (regression line). Describe the direction, strength, and form of the relationship observed.
34. Grouped Bar Chart or Stacked Bar Chart: Show how a numerical measure (mean or sum) varies across categories of two categorical variables simultaneously.
35. Correlation Heatmap: Compute the Pearson correlation matrix for all numerical variables and visualize it as an annotated heatmap. Identify the two strongest positive and two strongest negative correlations and comment on their plausibility.
36. Box-Violin Plot: Combine a violin plot with an overlaid boxplot for one numerical variable split by a categorical variable to show both distribution shape and summary statistics.

C3. Multivariate Analysis

37. Colour-encoded Scatter Plot: Create a scatter plot of two numerical variables with a third variable encoded as colour (hue). Add a fourth variable encoded as point size where meaningful. Describe the multivariate pattern observed.
38. Pair Plot (Seaborn pairplot or GGally ggpairs): Produce a pair plot for 4-6 key numerical variables with colour coding by a categorical variable. Identify one particularly interesting bivariate relationship revealed by this plot.
39. Faceted / Small-Multiples Plot: Create a grid of sub-plots showing the same relationship across different subgroups or categories (e.g., scatter plots faceted by region or gender). Comment on whether the relationship is consistent across subgroups.

C4. Temporal or Trend Analysis

40. If your dataset contains a date or time column: produce a line chart showing one or more numerical variables over time. Annotate any notable peaks, troughs, or trend changes.
41. If your dataset does not contain a temporal variable: produce a ranked or sorted bar chart that reveals a trend or ordering (e.g., top-10 countries by a metric, sales by product ranked), and annotate with data labels.

C5. Advanced or Creative Visualization

Produce one additional visualization of your choice that provides a meaningful and non-obvious insight from your dataset. This visualization should go beyond the standard charts above. Suitable options include (but are not limited to):

- Choropleth or geographic map (for datasets with country or regional data)
- Sankey or alluvial diagram (for flow or transition data)
- Treemap or sunburst chart (for hierarchical or proportional data)
- Radar / Spider chart (for multi-attribute comparisons)

- Animated or interactive Plotly / Altair figure
- Word cloud or frequency plot (for text-based datasets)
- Survival or cumulative distribution plot (for duration data)

Write a 3-4 sentence explanation of why you chose this visualization and what specific insight it reveals that simpler charts could not.

Part C-2: Data Analysis

All statistical tests must be performed using appropriate library functions. Do not compute test statistics manually unless specifically asked.

C6. Correlation Analysis

42. Compute Pearson correlation coefficients between all pairs of numerical variables and produce a ranked list of the top 5 strongest correlations (positive or negative).
43. For at least one pair, discuss whether the correlation is likely causal or merely associational, with a brief justification based on domain knowledge.
44. Compute a Spearman rank correlation for the same pairs and compare results. Note any differences and suggest what they imply about linearity or the presence of outliers.

C7. Key Patterns and Insights

45. Identify and articulate the three most significant patterns, relationships, or anomalies discovered across your entire analysis (Parts A-C). For each pattern: describe it clearly, reference the specific visualization or test that revealed it, and suggest a real-world implication or follow-up question.

Part D: Final Report and Code Quality

D1. Written Report

Write a structured analytical report documenting your findings. The report must be submitted as a separate PDF or Word document and must contain the following four sections:

46. Introduction (approx. 100 words): Introduce the dataset, describe where it comes from, and state the three main analytical questions you chose to investigate.
47. Methodology (approx. 150 words): Briefly describe the tools and methods used, cleaning decisions made, and any important preprocessing choices.
48. Key Findings (approx. 250-300 words): Summarize the most important findings from your visualizations and statistical tests. Reference at least four specific figures by number. Avoid repeating raw statistics already visible in the notebook; instead, interpret and contextualize them.
49. Conclusion and Limitations (approx. 150 words): Summarize what was learned from the data, discuss at least two limitations of the dataset or analysis, and suggest one direction for future work.

Embed at least four of your visualizations in the report with figure numbers and captions (e.g., "Figure 1: Distribution of passenger ages on the Titanic, grouped by survival status.").

D2. Code Quality and Reproducibility

50. All code cells must be clearly commented. Variable names must be descriptive. Avoid single-letter variable names except as loop counters.
51. Organize your notebook with Markdown section headers matching the Parts (A, B, C) of this examination.
52. The notebook must run completely from top to bottom using Kernel > Restart and Run All (Python) or knitr::knit() (R) without errors.
53. Include a requirements section at the top of the notebook listing all imported libraries with their versions.

Bonus Task: Interactive Dashboard (+5 marks)

NOTE: The bonus task is entirely optional. Marks earned here can supplement your total but cannot exceed 100. Attempting the bonus task will not negatively affect your score.

Build a simple interactive dashboard that allows a user to explore at least three visualizations from Part C dynamically. The dashboard must include at least one interactive filter or dropdown control. Acceptable frameworks:

- Python: Plotly Dash, Streamlit, Panel, or Voila
- R: Shiny
- Tableau

Submit the dashboard as: (a) a Python script (.py) or R Shiny app (app.R) in your zip archive, and (b) a URL to a deployed version (e.g., Streamlit Community Cloud, Shinyapps.io) if possible. A brief 2-3 sentence description of the dashboard's functionality must be included in your report.

4. Dataset Assignments by Student Number

Find your student number in the table below to identify your assigned dataset. The dataset number (DS01-DS32) must appear in your submission filename and on your cover sheet. All datasets are freely available from the listed sources. **Do not use any other dataset.**

#	Registration No.	Dataset	Domain	Source	Download URL / Reference
1	2025/HD05/26365U	Hotel Booking Demand Dataset	Economic	Kaggle	kaggle.com/datasets/jessemostipak/hotel-booking-demand
2	2025/HD05/26392U	World Happiness Report 2023	Social / Economics	Kaggle / UN SDG	kaggle.com/datasets/unsdsn/world-happiness
3	2025/HD05/26359U	Netflix Movies and TV Shows	Entertainment	Kaggle	kaggle.com/datasets/shivamb/netflix-shows
4	2025/HD05/26333U	COVID-19 Global Statistics	Public Health	Kaggle	kaggle.com/datasets/imdevskp/corona-virus-report
5	2025/HD05/26362U	Gapminder: Life Expectancy & GDP Per Capita	Global Development	Gapminder Foundation	gapminder.org/data
6	2024/HD05/25333U	Global Air Quality Index (2015-2023)	Environment	Kaggle / OpenAQ	kaggle.com/datasets/hasibalmustakim/air-quality-index-dataset
7	2025/HD05/26361U	FIFA World Cup Matches (1930-2022)	Sports	Kaggle	kaggle.com/datasets/abecklas/fifa-world-cup
8	2025/HD05/26353U	NYC Yellow Taxi Trip Records	Urban Mobility	NYC Open Data / TLC	nyc.gov/site/tlc/about/tlc-trip-record-data.page
9	2025/HD05/26350U	Global CO2 Emissions by Country	Environment	Kaggle	kaggle.com/datasets/ulrikthygpedersen/co2-emissions-by-country
10	2025/HD05/26355U	Student Performance in Secondary Education	Education	UCI ML Repository	archive.ics.uci.edu/dataset/320/student+performance
11	2025/HD05/26352U	Sample Superstore Retail Sales	Business / Retail	Kaggle / Tableau	kaggle.com/datasets/vivek468/superstore-dataset-final
12	2025/HD05/26368U	Avocado Prices in the USA (2015-2023)	Agriculture	Kaggle / Hass Avocado Board	kaggle.com/datasets/neuromusic/avocado-prices
13	2025/HD05/726343	US Baby Names by Year (2000-2019)	Demographics	Kaggle / SSA	kaggle.com/datasets/kaggle/us-baby-names
14	2025/HD05/26393U	Global Suicide Rates Overview (1985-2016)	Public Health	WHO / Kaggle	kaggle.com/datasets/russellyates88/suicide-rates-overview-1985-to-2016
15	2025/HD05/26347U	Forest Fires in Portugal (Montesinho Park)	Environment	UCI ML Repository	archive.ics.uci.edu/dataset/162/forest+fires
16	2025/HD05/26354U	Cleveland Heart Disease Dataset	Healthcare	UCI ML Repository	archive.ics.uci.edu/dataset/45/heart+disease
17	2500726334	Pima Indians Diabetes Dataset	Healthcare	UCI ML Repository / Kaggle	kaggle.com/datasets/uciml/pima-indians-diabetes-database
18	2025/HD05/26339U	Wine Quality Dataset (Red & White)	Food & Beverage	UCI ML Repository	archive.ics.uci.edu/dataset/186/wine+quality
19	2500726356	Airbnb Listings in New York City	Urban / Tourism	Inside Airbnb	insideairbnb.com/get-the-data
20	2025/HD05/26336U	Olympic Athletes and Events (1896-2016)	Sports / History	Kaggle	kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results
21	2025/HD05/26358U	Global Food Security & Nutrition Data	Agriculture / Health	FAO / Kaggle	kaggle.com/datasets/ruchi798/food-choices-and-nutritional-facts

#	Registration No.	Dataset	Domain	Source	Download URL / Reference
22	2024/HD05/21952U	Video Game Sales (1980-2020)	Entertainment	Kaggle	kaggle.com/datasets/gregorut/videogamesales
23	2025/HD05/26394U	Spotify Top Songs by Year (2010-2023)	Music / Media	Kaggle / Spotify API	kaggle.com/datasets/leonardopena/top-spotify-songs-from-20102019-by-year
24	2025/HD05/26332U	Global Renewable Energy Statistics	Energy	Kaggle	kaggle.com/datasets/belayethossainds/renewable-energy-world-wide-19652022
25	2025/HD05/26374U	Online Retail Transactions (E-commerce)	Business / Commerce	UCI ML Repository	archive.ics.uci.edu/dataset/352/online+retail
26	2025/HD05/26501U	World Population Prospects by Country	Demographics	UN Population Division	kaggle.com/datasets/iamsouravbanerjee/world-population-dataset
27	2025/HD05/26371	Global Education Attainment Statistics	Education	World Bank EdStats	databank.worldbank.org/source/education-statistics
28	2025/HD05/26357U	US Gun Violence Incidents (2013-2018)	Social / Policy	Kaggle / Gun Violence Archive	kaggle.com/datasets/jameslko/gun-violence-data
29	2025/HD05/26373U	Chicago Crimes Dataset (2010-2023)	Public Safety	Chicago Data Portal	data.cityofchicago.org/Public-Safety/Crimes
30	2025/HD05/26375U	Global Malaria Incidence and Deaths	Public Health	Kaggle	kaggle.com/datasets/imdevskp/malaria-dataset
31	2025/HD05/31216U	Global Plastic Waste Production by Country	Environment	Kaggle	kaggle.com/datasets/sohamgade/plastic-pollution-data
32	2024/HD05/21933U	Amazon Product Ratings and Reviews	E-commerce / NLP	Kaggle / UCSD	kaggle.com/datasets/saurav9786/amazon-product-reviews
33	2025/HD05/26342U	IBM HR Analytics Employee Attrition	Business	Kaggle	kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
34	2025/HD05/26397U	Global Daily Weather and Climate Records	Meteorology / Environment	Kaggle / NOAA	kaggle.com/datasets/guillemservera/global-daily-climate-data

End of Examination